BEGINNER'S STUDY GUIDE · END TO END

Transformers in NLP: Understand the Project, Present It with Confidence

From “what is sentiment analysis?” to “how do I answer the professor?”, with real-world examples, the real numbers of our experiment, a paper map, a slide-by-slide speaking guide and tips for the questions.

The project in one sentence

We compared ten ways of classifying movie-review sentences as positive or negative, from simple word counting to Transformers, on one ordinary laptop, and measured not only *how accurate* each is but also *how much it costs* and *how many labelled examples it needs*.

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All numbers come from the project's real results (SST-2 test set, 872 sentences, one laptop CPU). Companion material: paper (25 pages), presentation (11 slides), interactive study guide, project map with 54 interview questions.

How to use this guide

If you have 1 day
Read Part 2 (the project end to end) and Part 4 (explaining to the professor). Skim the concept boxes you do not know. Rehearse the slides once with a timer.

If you have 3 days
Day 1: Part 1 concepts. Day 2: Part 2 and the paper map. Day 3: slide guide, questions, two full rehearsals.

If you have a week
Do the 3-day plan, then add the interactive study guide (test your own sentences on all ten models), the project map quiz mode, and a mock viva with a friend.

Every concept box has three parts

IN PLAIN WORDS
The idea with no jargon, as you would explain it to a friend.

REAL-WORLD EXAMPLE
Something from daily life (food-delivery reviews, a bus pass counter, cricket scores) that makes the idea stick.

WHERE IT APPEARS IN OUR PROJECT
Exactly where this idea shows up in our experiment, paper or slides, with the real numbers.

The project on one page



Question we asked	How we answered it	Short answer
Do Transformers really beat older methods?	Ten models, same data, same laptop	Yes: fine-tuned DistilBERT 88.8% vs best classical 82.9%
What does the improvement cost?	Measured training time, speed, size	20 to 2,000 times slower per sentence; DistilBERT needs 2.6 h to fine-tune on the CPU
How much labelled data is needed?	Learning curve, 500 to 8,000 labels	With 500 labels BERT-small 75.9% vs TF-IDF 61.8%
Is reusing a public model enough?	Three ready-made models	Best: 91.1% with no training at all
Are Transformers better on tricky sentences?	Negation and contrast subsets	TF-IDF + SVM drops 8.4 points on negation, DistilBERT-SST2 4.1

THE MOTTO OF THE PROJECT
Explain it. Measure it. Make it useful for everyone. We explain how Transformers work, we measure honestly when they are worth their cost, and we turn what we learned into material others can use.

PART 1 · CONCEPTS FROM ZERO

1. The problem: sentiment analysis

What is sentiment analysis?

IN PLAIN WORDS

Teaching a computer to decide whether a piece of text is positive or negative. It is a **classification** task: the answer is one of a few labels.

REAL-WORLD EXAMPLE

A restaurant on a food-delivery app gets 5,000 reviews a month. Nobody can read them all. A sentiment model reads every review and reports "78% positive, 22% negative", and flags the negative ones about delivery delays.

WHERE IT APPEARS IN OUR PROJECT

Our task is exactly this, with movie-review sentences. Each sentence is labelled **1 (positive)** or **0 (negative)**. The data set is called **SST-2** (Stanford Sentiment Treebank, two classes).

Why is it harder than it looks?

Sentence	A word-counting model thinks	The truth	Why it is hard
The biryani was great.	positive	positive	easy: one positive word
The biryani was **not** great.	positive (sees "great")	negative	**negation** flips the meaning
The food was good, **but** the delivery was terribly late.	mixed	negative overall	**contrast**: the part after "but" decides
Oh great, another two-hour wait.	positive (sees "great")	negative	**sarcasm**: needs real understanding

WHERE IT APPEARS IN OUR PROJECT

We specifically measure accuracy on **184 test sentences with negation** (not, no, never, n't ...) and **130 with contrast** (but, although, though, yet, however). This is research question RQ5.

The data: SST-2

Movie-review sentences from Rotten Tomatoes. Think of three kinds of exam papers:

Part	Size	Real-world picture	Used for
Training set	65,349 sentences	practice papers	the model learns from these
Development set	2,000 sentences	mock exam	choosing the best epoch (when to stop)
Test set	872 sentences	the real exam: questions never seen	the final honest score; never used for learning

Training sentences are short phrases (about 9.4 words on average); test sentences are full sentences (about 19.5 words). So models must generalise from phrases to whole sentences, which makes the task a fair challenge.

2. From text to numbers, and the classical methods

Tokens, vocabulary and ids

IN PLAIN WORDS

A computer cannot read letters. We cut a sentence into pieces called **tokens** (mostly words) and give every token a number from a fixed list called the **vocabulary**.

REAL-WORLD EXAMPLE

A bus-pass counter writes every stop name in a register with a serial number: Majestic = 1, Mysuru Road = 2 … Staff then talk in numbers. Words that are not in the register are "unknown".

WHERE IT APPEARS IN OUR PROJECT

Classical models use words and word pairs as features. The Transformers use **WordPiece** pieces: "unwatchable" becomes *un ##watch ##able*, so even unseen words can be understood from their pieces.

Bag of words and TF-IDF

IN PLAIN WORDS

Bag of words counts how often each word occurs and forgets the order. **TF-IDF** improves this: common words like "the" get a low weight, rare informative words get a high weight.

REAL-WORLD EXAMPLE

Three delivery reviews: (1) "biryani was great", (2) "delivery was late", (3) "the delivery was great". The word "was" appears in all three, so it tells us nothing. "biryani" appears once, so it is informative.

WHERE IT APPEARS IN OUR PROJECT

All three of our classical models use TF-IDF features (about 73,000 word and word-pair weights).

Word	Reviews containing it (of 3)	IDF = ln((1+3)/(1+df)) + 1	Meaning
was	3	1.00	common: low weight
delivery	2	1.29	medium
biryani	1	1.69	rare: high weight

Naive Bayes, logistic regression, linear SVM

IN PLAIN WORDS

Three simple ways to turn word weights into a decision. Each word votes a little for "positive" or "negative", and the votes are added up.

REAL-WORLD EXAMPLE

A panel of judges where each judge only knows a few words: "great" gets +2 votes, "late" gets −2. The sum decides. Quick and cheap, but the judges ignore how words combine.

WHERE IT APPEARS IN OUR PROJECT

Our best classical model, TF-IDF + linear SVM, reaches **82.9%**, trains in 2 seconds and classifies a sentence in 0.06 ms. It is the strong, cheap baseline every other model must beat.

THEIR WEAKNESS

Because they ignore word order, "not great" still contains "great". On our negation sentences the SVM drops from 82.9% to 74.5%.

3. Neural networks, reading in order, and the Transformer

How a neural network learns

IN PLAIN WORDS

A network is a long chain of simple calculations with many adjustable numbers (**weights**). Learning is a loop: guess, measure how wrong the guess was (**loss**), work out which weights to change (**backpropagation**), change them a little (**gradient descent**), repeat thousands of times.

REAL-WORLD EXAMPLE

A cricket coach tells a batter: "your shot was too early". The batter adjusts a little and tries again. After thousands of balls the timing improves. The weights are the batter's habits; the loss is the coach's complaint.

WHERE IT APPEARS IN OUR PROJECT

We fine-tune three Transformers with an optimiser called **AdamW** and choose the best epoch on the development set.

Reading in order: RNN, LSTM, BiLSTM

IN PLAIN WORDS

A recurrent network reads one word at a time and carries a memory forward. An **LSTM** adds gates so it can keep important things longer. A **BiLSTM** reads the sentence both forwards and backwards.

REAL-WORLD EXAMPLE

Reading a long WhatsApp message aloud and trying to remember its start by the time you reach the end. The longer it is, the more you forget, and you cannot start the last line before finishing the first.

WHERE IT APPEARS IN OUR PROJECT

Our BiLSTM (2.0 M parameters, trained from scratch) reaches 82.8%, almost the same as the SVM (82.9%). It is better on negation (78.8% vs 74.5%) because it reads word order, but without pre-training it brings no overall gain.

The Transformer and self-attention

IN PLAIN WORDS

The Transformer (Vaswani et al., 2017) does not read word by word. Every word looks at **all** the other words at once and decides which ones matter for understanding it. This "looking" is called **self-attention**. Because all words are processed together, training is fast and the model can be made very large.

REAL-WORLD EXAMPLE

A group discussion: when one person speaks, everyone listens, but they pay most attention to the people whose points relate to theirs. In "the plot was **not good**, but the acting was **brilliant**", the word "good" pays attention to "not" and the word "brilliant" attends to "acting".

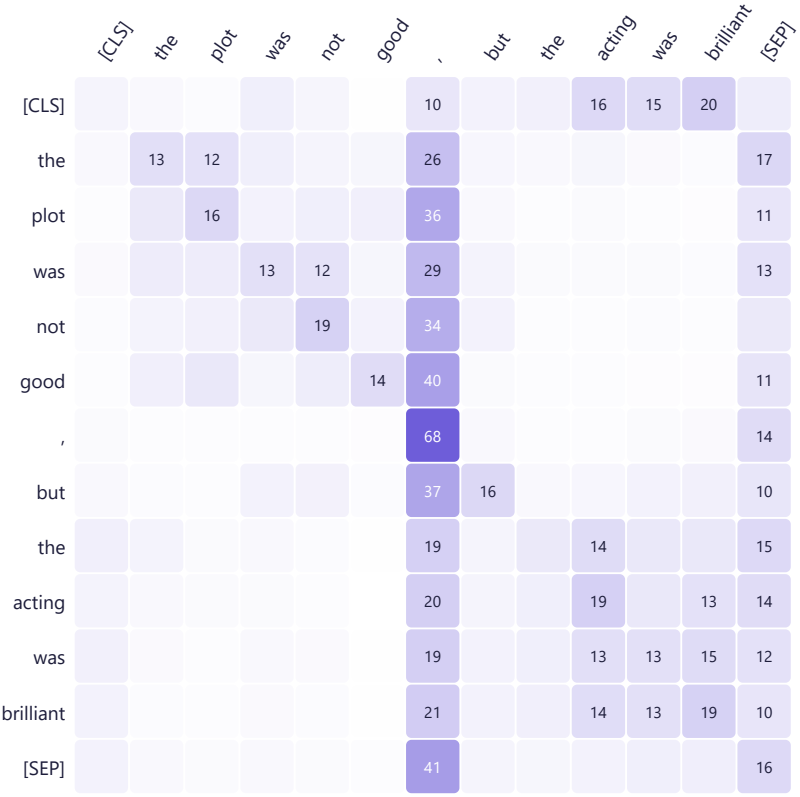
WHERE IT APPEARS IN OUR PROJECT

Three of our models are Transformer *encoders* (BERT-tiny, BERT-small, DistilBERT). For classification we take the special **[CLS]** vector, which summarises the whole sentence, and add one layer that outputs positive or negative. The formula on slide 3 is **softmax(QK^T/√d)·V**: Q = what each word is looking for, K = what each word offers, V = the content it passes on.

PART 1 · CONCEPTS FROM ZERO

4. What attention looks like, and what to say about it

This map shows the **real attention** of the DistilBERT model (last layer, averaged over its heads) for the sentence “the plot was not good, but the acting was brilliant”. Each row is a word that is looking; each column is a word it looks at; numbers are percent and each row adds up to 100.



Real attention from DistilBERT-SST2 (the same sentence as slide 10). Cell darkness = share of attention.

HOW TO TALK ABOUT THIS FIGURE HONESTLY (IMPORTANT IN THE VIVA)

Notice that a lot of attention goes to the comma and the [SEP] marker, which act as “rest” positions, and that some words attend to their neighbours (for example “acting” and “brilliant”). Say: “The map shows where information *can* flow between words.” Do **not** say: “this proves the model understands negation”. Jain and Wallace (2019) warned that attention weights are an illustration, not a proof of reasoning. Our slide uses the same wording on purpose.

Pre-training, fine-tuning, zero-shot and ready-made models

IN PLAIN WORDS

Pre-training: the model reads a huge amount of text once (for BERT, by guessing hidden words) and learns English in general. **Fine-tuning**: we then train it a little on our labelled sentences so it learns our task. **Ready-made**: someone else has already fine-tuned a model and we simply use it. **Zero-shot**: a model never trained for sentiment is asked to decide anyway.

REAL-WORLD EXAMPLE

A fresh graduate who has studied for years (pre-training) joins a company and learns its processes in a few weeks (fine-tuning). A ready-made model is an experienced colleague you borrow. A zero-shot model is a smart person doing a task for the first time with only the instructions.

WHERE IT APPEARS IN OUR PROJECT

Fine-tuned by us on 8,000 sentences: BERT-tiny, BERT-small, DistilBERT. Ready-made: DistilBERT-SST2 (91.1%), Twitter-RoBERTa (85.8%, a different domain), and a zero-shot NLI model (81.8% with **no** labelled examples).

5. How to judge models: the measurements

Measurement	Plain meaning	Real-world picture	Our use
Accuracy	share of test sentences classified correctly	marks in the exam out of 872	main quality number
Macro-F1	accuracy that treats both classes equally	marks that cannot be gamed by always answering the common class	checked alongside accuracy (nearly identical here)
Training time	seconds or hours to build the model	how long the student studies	RQ2: cost
Latency	milliseconds to classify one sentence	how long the counter clerk takes per customer	RQ2: cost of using the model
Parameters	number of adjustable numbers	size of the clerk's rulebook	memory and size
Negation / contrast accuracy	accuracy only on the tricky sentences	marks on the hardest questions	RQ5
Learning curve	accuracy as the number of labels grows	marks after 10, 50, 100 practice papers	RQ3: data efficiency

Confidence interval and McNemar's test

IN PLAIN WORDS

A test set is only a sample. If we had picked 872 different sentences the accuracy would be slightly different. The **95% confidence interval** is a range that shows how much it could wobble. **McNemar's test** compares two models on the *same* sentences and asks: is one really better, or is the difference luck?

REAL-WORLD EXAMPLE

Two batters average 41 and 43 over 10 matches. Is the second one truly better? With so few matches you cannot tell. Over 200 matches you could. The interval is the batting average with an honest “plus or minus”.

WHERE IT APPEARS IN OUR PROJECT

We resampled the test set many times (bootstrap) to get an interval for every model; for example DistilBERT-SST2 91.1%, interval 89.1 to 92.9. McNemar's test shows our fine-tuned DistilBERT is significantly better than the best classical model ($p < 0.001$).

RULE OF THUMB FOR 872 TEST SENTENCES

The wobble of an accuracy near 85% is about ±2.4 points. Treat differences under 2 to 3 points between single models as *not clearly real*; larger gaps (88.8% vs 82.9%) are clearly real.

6. The gap and the five research questions

The gap in one paragraph. Studies of Lithuanian sentiment classification show that pre-trained models (LitLat BERT, multilingual BERT and T5, Lithuanian Llama2) work, and that neural networks without pre-training do not beat classical methods on small data (naive Bayes 0.735 against CNN 0.706). But they report accuracy; they seldom measure cost (time, speed, size), how many labels a pre-trained model needs, or compare ready-made models with a fine-tuned in-domain model under one protocol. Our project measures all of these together.

REAL-WORLD EXAMPLE OF THE GAP

A small company in Vilnius must choose a sentiment model for customer reviews and has one laptop. The papers tell it "BERT works". They do not tell it how long it will take on a laptop, how many reviews it must label first, or whether it can just download a ready-made model. We answer those questions.

RQ	Question	What we measured	Where in the paper
RQ1	How much more accurate are Transformers?	accuracy and macro-F1 of ten models	Chapter 3, quality section
RQ2	What does the improvement cost?	training time, latency, parameters	Chapter 3, cost section
RQ3	How does the advantage depend on labels?	learning curve 500 to 8,000 labels	Chapter 3, data efficiency
RQ4	Ready-made vs trained by us?	three ready-made models vs fine-tuned ones	Chapter 3, ready-made models
RQ5	Better on negation and contrast?	accuracy on 184 + 130 tricky sentences	Chapter 3, linguistic subsets

The ten competitors

Model	Family	Parameters	Training labels	Accuracy	Training time	Latency
TF-IDF + naive Bayes	Classical	147 k	65,349	82.2%	2 s	0.05 ms
TF-IDF + logistic reg.	Classical	74 k	65,349	81.9%	4 s	0.05 ms
TF-IDF + linear SVM	Classical	74 k	65,349	82.9%	2 s	0.06 ms
BiLSTM (scratch)	Recurrent	2.0 M	65,349	82.8%	6 min	0.45 ms
BERT-tiny (fine-tuned)	Transformer	4.4 M	8,000	76.6%	2 min	1.17 ms
BERT-small (fine-tuned)	Transformer	28.8 M	8,000	83.7%	26 min	14.11 ms
DistilBERT (fine-tuned)	Transformer	67.0 M	8,000	88.8%	2.6 h	39.13 ms
DistilBERT-SST2 (ready)	Transformer	67.0 M	0	91.1%	none	44.56 ms
Twitter-RoBERTa (ready)	Transformer	124.6 M	0	85.8%	none	84.80 ms
Zero-shot NLI	Transformer	82.1 M	0	81.8%	none	107.02 ms

Parameters for the TF-IDF models are the number of word weights (73 k to 147 k), not a neural network. Latency = time to classify one sentence (batches of 32).

PART 2 · THE PROJECT END TO END

7. The pipeline: what each script does

The program is nine small Python scripts that pass files to each other. You can explain it as “data in, ten models, results out, report”.

Script	What it does (plain words)	What it produces
01_prepare_data.py	Downloads SST-2 and splits it into training, development and test sentences	train.csv, dev.csv, test.csv
02_classical_baselines.py	Turns sentences into TF-IDF numbers and trains naive Bayes, logistic regression, linear SVM	3 result files
03_bilstm.py	Builds a vocabulary and trains the BiLSTM from scratch	1 result file
04_finetune_transformer.py	Takes any pre-trained model name and a number of labels, adds a classification layer and fine-tunes it (AdamW, warm-up, linear decay)	BERT-tiny, BERT-small, DistilBERT results + learning-curve runs
05_pretrained_transformers.py	Runs the three ready-made models on the test sentences with no training	3 result files
06_learning_curve_classical.py	Repeats the 500 to 8,000 label experiment for TF-IDF	learning_curve.csv
07_attention_visualisation.py	Extracts attention weights of a fine-tuned model for an example sentence	attention figure
08_report.py	Collects all result files; makes tables, figures, subset analysis, confidence intervals and McNemar tests	tables and charts for the paper
09_diagrams.py	Draws the pipeline and architecture diagrams	figures

WHY THIS DESIGN IS GOOD (SAY IT TO THE PROFESSOR)

Every model writes the **same kind of result file** including its individual predictions. That makes the study reproducible and lets us compute paired statistics afterwards without retraining. A single command, `run_all.sh`, reproduces everything in about 2.5 to 3 hours on the laptop.

The set-up that makes the comparison fair

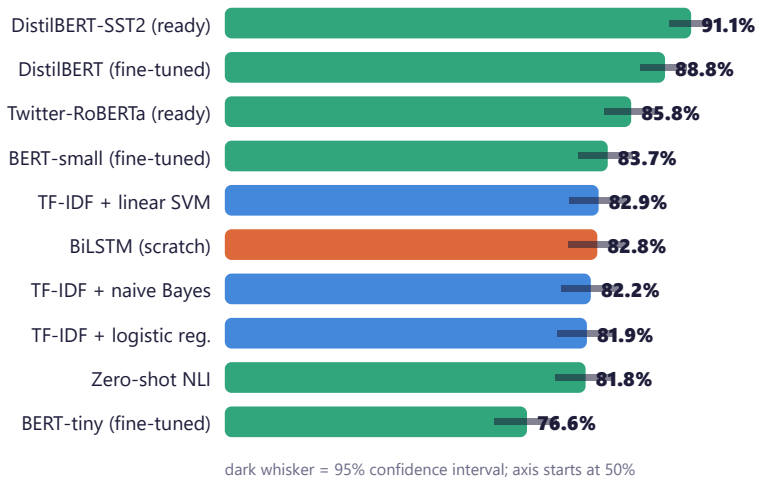
Fixed for all models	Value
Computer	Intel Core i5-8365U laptop, 4 cores, 16 GB RAM, **no GPU**
Data split	same train / development / test sentences for every model
Random seed	42 for splits, sampling and initialisation
Fine-tuning labels	8,000 sentences for each of the three fine-tuned Transformers
Test set	the 872 official validation sentences, never used for learning or choosing settings

REAL-WORLD EXAMPLE

A fair race: same track, same day, same stopwatch. If one runner had a bicycle the times would mean nothing. Running everything on one CPU and one split is our “same track”.

8. The results, one question at a time

RQ1 · Accuracy

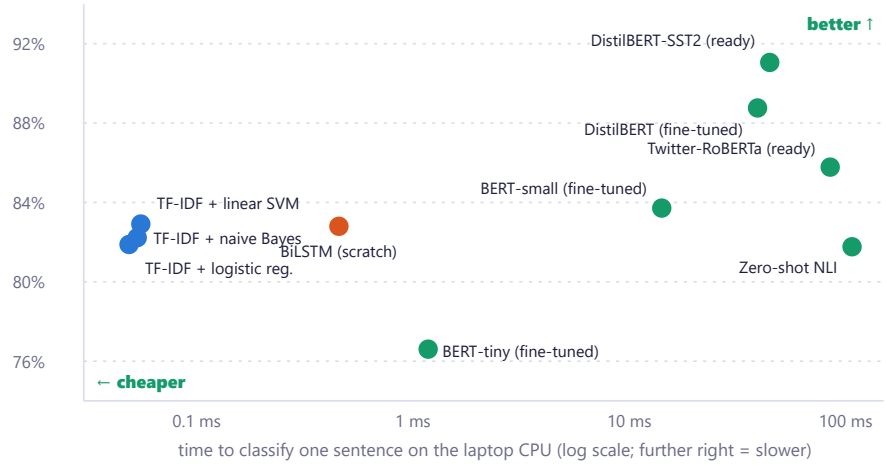


Plain meaning: the ready-made DistilBERT is best (91.1%). Our DistilBERT, fine-tuned on only 8,000 sentences, reaches 88.8%, which is 5.8 points above the best classical model although that model saw 65,349 sentences. The BiLSTM (82.8%) gives no advantage over the SVM. Tiny Transformers are not enough: BERT-tiny is the weakest model (76.6%).

WHAT TO SAY

"A Transformer fine-tuned on eight times less data beat the best classical model, and the difference is statistically significant. Pre-training is the reason."

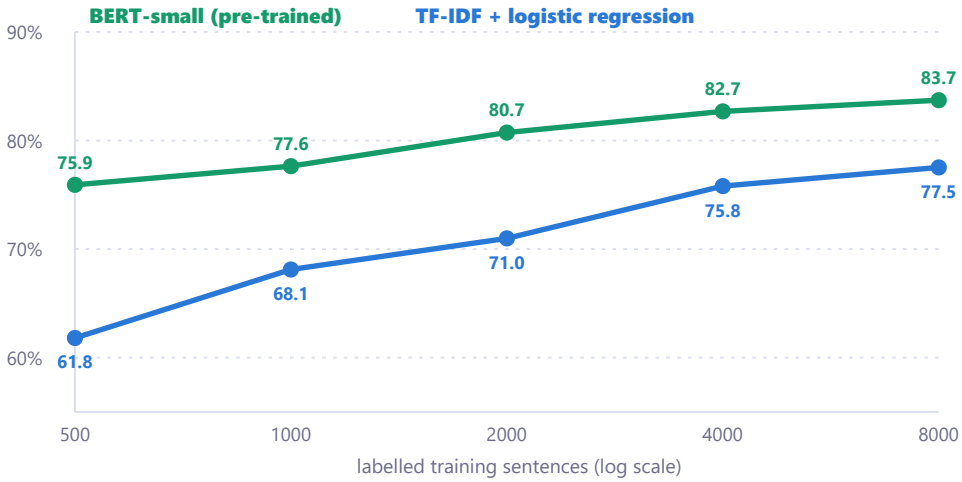
RQ2 · Cost: the accuracy–speed trade-off



Plain meaning: the best models are on the right (slow). TF-IDF answers in 0.06 ms; DistilBERT in about 39 to 45 ms; the zero-shot NLI model in 107 ms. That is 20 to 2,000 times slower. Fine-tuning DistilBERT on the CPU took 2.6 h; the SVM took 2 seconds. For most real uses, 40 ms per sentence is still perfectly fast, so the extra cost is usually acceptable.

8. The results (continued)

RQ3 · Data efficiency

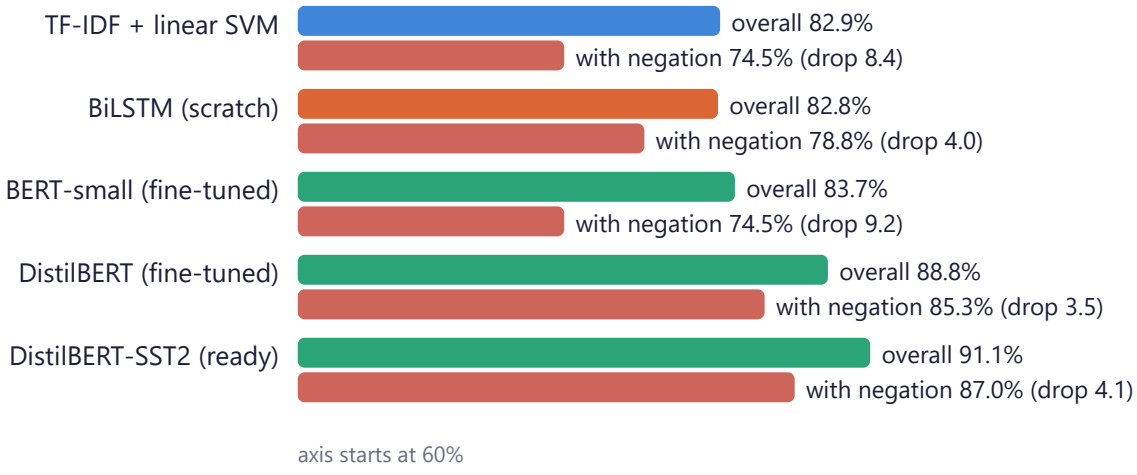


Plain meaning: with only 500 labelled sentences BERT-small scores 75.9% against 61.8% for TF-IDF: **+14.1 points**. With 4,000 labels it already matches the classical model trained on 65,349 (82.7% vs 82.9%). The gap shrinks as labels grow, but it never disappears in the range we tested. **Why:** BERT-small already knows English from pre-training; TF-IDF must learn every word from our labels (with 500 labels, about 29% of the words in the test sentences were never seen).

RQ4 · Ready-made models

Plain meaning: the cheapest way to a good model is to reuse one. DistilBERT-SST2 gives 91.1% with zero training. Twitter-RoBERTa (85.8%), trained on tweets, still works well on movie reviews but loses a little because of the domain change. The zero-shot NLI model (81.8%) uses **no** labelled data at all and still matches classical models, though it is the slowest.

RQ5 · Negation and contrast



Plain meaning: bag-of-words models lose 8 to 9 points on sentences with negation; the large Transformers lose only 2 to 4 (DistilBERT-SST2: 91.1% overall, 87.0% on negation; SVM: 82.9% and 74.5%). Self-attention lets "not" modify "good" even when the words are far apart.

HONEST NOTE

The small fine-tuned models (BERT-tiny, BERT-small) are *not* better on tricky sentences. The advantage on negation belongs to the larger Transformers.

9. One sentence through all ten models

THE SENTENCE (TRUE LABEL: NEGATIVE)

“a great ensemble cast ca n't lift this heartfelt enterprise out of the familiar .”

A word-counting model sees *great* and *heartfelt* (positive words) and misses that “can't lift … out of the familiar” makes it negative.

Model	Predicts	
TF-IDF + naive Bayes	positive	✗
TF-IDF + logistic reg.	positive	✗
TF-IDF + linear SVM	positive	✗
BiLSTM (scratch)	positive	✗
BERT-tiny (fine-tuned)	positive	✗
BERT-small (fine-tuned)	positive	✗
DistilBERT (fine-tuned)	negative	✓
DistilBERT-SST2 (ready)	negative	✓
Twitter-RoBERTa (ready)	negative	✓
Zero-shot NLI	negative	✓

WHAT THIS SHOWS

The three TF-IDF models and the BiLSTM, which learn only from our labelled sentences, call it positive. The larger Transformers and the zero-shot model get it right: from pre-training they already know how “can't lift” and “familiar” behave in English. The two small fine-tuned models have less capacity to use that knowledge and also miss it.

USE IT IN THE TALK

One real sentence is worth a page of numbers. Show it before the results slide: “Here is why accuracy differs.”

Confidence intervals and tricky-sentence accuracy for every model

Model	Accuracy	95% interval	Negation (184)	Contrast (130)
DistilBERT-SST2 (ready)	91.1%	89.1 to 92.9	87.0%	86.2%
DistilBERT (fine-tuned)	88.8%	86.6 to 90.7	85.3%	85.4%
Twitter-RoBERTa (ready)	85.8%	83.4 to 88.1	83.7%	84.6%
BERT-small (fine-tuned)	83.7%	81.1 to 85.9	74.5%	76.2%
TF-IDF + linear SVM	82.9%	80.4 to 85.3	74.5%	77.7%
BiLSTM (scratch)	82.8%	80.3 to 85.2	78.8%	77.7%
TF-IDF + naive Bayes	82.2%	79.6 to 84.7	73.9%	76.9%
TF-IDF + logistic reg.	81.9%	79.2 to 84.4	73.9%	76.9%
Zero-shot NLI	81.8%	79.1 to 84.2	81.5%	79.2%
BERT-tiny (fine-tuned)	76.6%	73.7 to 79.2	69.6%	65.4%

Intervals of the top models overlap (for example 86.6 to 90.7 for fine-tuned DistilBERT and 89.1 to 92.9 for DistilBERT-SST2), so we do not claim that the ready-made model is clearly better than our fine-tuned one.

10. Conclusions, recommendations and limits

What we found

1. Fine-tuned Transformers are the most accurate: 88.8% against 82.9% (classical) and 82.8% (BiLSTM).
2. Pre-training gives data efficiency: BERT-small with 4,000 labels matches classical models trained on 65 k.
3. The cost is real but manageable: minutes to hours of CPU fine-tuning, milliseconds per sentence.
4. Reuse first: a ready-made in-domain model gives 91.1% with no training.
5. Large Transformers handle negation and contrast much better.

What we recommend

1. Look for a public model already fine-tuned for your task and domain.
2. If none exists, fine-tune a distilled Transformer such as DistilBERT on a few thousand labels. This works even without a GPU.
3. With no labels at all, a zero-shot model is a usable start.
4. Keep TF-IDF only when inference must be extremely cheap or the hardware is very limited.

Limits we state openly (this builds trust with the professor)

Limit	Why it matters	What we would do
One dataset (English movie reviews)	results may differ for other domains and languages	repeat on other datasets, Lithuanian and Kannada text
One random seed	run-to-run variation is not measured; we use bootstrap intervals instead	repeat with several seeds and report the spread
872 test sentences	differences under 2 to 3 points are not clearly real	use bigger test sets and paired tests for close pairs
CPU-only timing	GPU timing would change the ratios	measure on a GPU as well
8,000 fine-tuning labels	other amounts could change the ranking of small models	extend the learning curve to the fine-tuned DistilBERT

FUTURE WORK (ONE SENTENCE FOR THE CLOSING SLIDE)

The most natural continuation is the same procedure for Lithuanian texts with LitLat BERT or XLM-RoBERTa, with several random seeds and GPU timing.

11. The paper, chapter by chapter

The paper has 25 pages in the VILNIUS TECH template. Use this map to find things quickly and to know what each part is for.

Part	Pages	What is inside	If the professor asks...
Title page, contents	1 to 3	title, authors, table of contents, list of abbreviations	—
Introduction	4 to 5	why the topic matters, research aim, objectives, methods, work distribution (two students)	"What is the aim?": to analyse the Transformer and evaluate it experimentally against classical and recurrent methods for sentiment classification.
1. Literature review	6 to 8	search strategy (Google Scholar, Web of Science), history from word vectors to Transformers, model families, Lithuanian research, summary table, the gap	"What did you read and what is missing?" Use the gap paragraph from section 6.
2. Research methodology	9 to 16	problem, dataset, ten models, criteria, program design, **the Transformer architecture** (tokenisation, positional encoding, attention, multi-head, feed-forward, complexity), mathematics of training and evaluation, protocol and threats to validity	"Explain attention." Section 2.6, pages 12 to 14. "How did you evaluate?" Section 2.7, pages 14 to 16.
3. Practical part	17 to 22	implementation, quality results, cost, data efficiency, ready-made models, negation and contrast, comparison with previous research	"What did you find?" Use section 8 of this guide.
Conclusions	23	five numbered conclusions and practical recommendations	"What do you recommend?"
References	24 to 25	38 entries	"Where is this from?"

HOW TO READ THE PAPER IN 30 MINUTES

1) Introduction (aim and objectives). 2) The summary table in the literature review and its conclusions. 3) Sections 2.2 and 2.3 (data and models). 4) Section 2.6 headings only. 5) The results tables and the five conclusions. Then read the slides next to them: each slide matches one part of the paper.

NUMBERS YOU MUST KNOW BY HEART

872 test sentences · 65,349 training · 10 models · 4 families · 8,000 fine-tuning labels · best 91.1% (ready-made) · 88.8% (ours) · 82.9% (best classical) · +14.1 points at 500 labels · 20 to 2,000× slower · laptop i5-8365U, no GPU.

12. The presentation, slide by slide

11 slides, about **9 minutes** in total (Shashikanth about 4 min 40 s, Shaeed about 4 min 20 s). Practise saying the quoted text in your own words, not word for word.

#	Speaker, time	Goal of the slide	What to say (sample)	Key number	Likely question	Tip
1	Shashikanth 30 s	Title: introduce yourselves, the topic, and the question.	"Transformers power BERT, GPT and almost every modern NLP system. Our question: what do they really give, and at what cost, on an ordinary laptop with limited labelled data?"	The question (not a number).	Why a laptop?	Smile, state the question slowly; do not rush the opening.
2	Shashikanth 45 s	Show the shift from reading word by word to reading all at once.	"Before 2017 models read a sentence word by word. The Transformer removed recurrence and kept only attention: every word looks at every other word, all at the same time. That allowed huge pre-training: BERT, GPT."	2017, attention only.	What is the difference from an LSTM?	Use the analogy: a group discussion instead of passing a message down a line.
3	Shashikanth 60 s	Explain self-attention and the architecture simply.	"Each word asks a question (query), every word offers a label (key) and content (value). The softmax of the dot products gives weights; the output is a weighted mix of the values. Several heads run in parallel, layers are stacked. For classification we take the [CLS] vector."	$\text{softmax}(QK^T/\sqrt{d})\cdot V$.	Why divide by $\sqrt{d}$?	Point at Q, K, V on the slide as you say them; do not read the formula.
4	Shaeed 60 s	Literature: what is known and what is missing, in three plain sentences plus the gap.	"In one sentence: the papers tell us which model is accurate, but not what it costs or how many labels it needs. Three things are known: Transformers are most accurate; without pre-training small data is a problem, as the Lithuanian study showed; and Lithuanian now has pre-trained models. What is missing is the cost, the number of labels and whether a ready-made model is enough. We measure those on one laptop."	NB 0.735 vs CNN 0.706; gap = cost + labels + ready-made.	What exactly is the gap?	Say the gap as one sentence, then pause: "accuracy is reported, cost and label needs are not."

12. The presentation (continued)

#	Speaker, time	Goal of the slide	What to say (sample)	Key number	Likely question	Tip
5	Shaeed 45 s	The experiment design.	"SST-2 movie reviews. The official validation set is our test set; 2,000 training sentences are the development set. Ten models in four groups, all on the same laptop CPU so the times are comparable."	65k / 2k / 872; 10 models.	Why is the validation set your test set?	Explain train, development and test with the exam analogy.
6	Shashikanth 40 s	Implementation: a reproducible pipeline.	"Nine scripts, one generic fine-tuning script that takes any model name from the Hugging Face hub. Every model saves metrics, timings and predictions, so the report script computes tables and statistics. run_all.sh reproduces everything."	~2.5 h to reproduce.	How reproducible is it?	Mention the fixed seed (42) and saved predictions.
7	Shashikanth 60 s	Result 1: accuracy.	"Classical models reach about 82 to 83%, the BiLSTM 82.8%. Our fine-tuned DistilBERT reaches 88.8% with only 8,000 labels. The ready-made DistilBERT-SST2 reaches 91.1%. Zero-shot, with no labels, still gets 81.8%."	88.8% vs 82.9%.	Is 88.8 vs 91.1 a real difference?	Say the three big numbers once, then stop talking and let them look.
8	Shashikanth 45 s	Result 2: the price.	"The gain is not free: TF-IDF trains in seconds and answers in 0.06 milliseconds; the Transformers need minutes to hours of CPU training and milliseconds per sentence. For most applications that is still acceptable."	20 to 2,000× slower.	Is the cost worth it?	Read one row of the table (SVM vs DistilBERT) instead of the whole table.

12. The presentation (continued)

#	Speaker, time	Goal of the slide	What to say (sample)	Key number	Likely question	Tip
9	Shaeed 50 s	Result 3: data efficiency.	"With only 500 labels BERT-small is 14 points better than TF-IDF, because it already knows English from pre-training. With 4,000 labels it matches the classical model trained on 65,000. This agrees with the Lithuanian finding that pre-training, not the neural network as such, matters on small data."	+14 points at 500 labels.	Why does small data hurt?	Use the "graduate vs fresher" analogy for pre-training.
10	Shaeed 45 s	Result 4: negation and contrast.	"Bag-of-words models lose the most on sentences with negation and contrast, because 'not good' contains 'good'. Transformers keep their accuracy much better. The heat map shows how attention links the words, but it is an illustration, not a proof of reasoning."	8 to 9 vs 2 to 4 points lost.	Does attention explain the decision?	Use the real example sentence (section 9 of this guide) before the map.
11	Shaeed 60 s	Conclusions and recommendation.	"Fine-tuned Transformers are most accurate; pre-training gives data efficiency; the cost is real but manageable. Recommendation: first try a public in-domain model; otherwise fine-tune a compact Transformer on a few thousand labels; keep TF-IDF only when latency or hardware are extremely limited. Next: Lithuanian data with LitLat BERT."	91.1% with zero training.	What would you do differently?	Finish with the recommendation, then say "thank you" and invite questions.

Handover lines between the two speakers

After slide	From → to	Say
3	Shashikanth → Shaeed	"That is how the architecture works. Shaeed will now show what the literature says and what is missing."
5	Shaeed → Shashikanth	"Now Shashikanth will explain how we implemented the experiment and show the results."
8	Shashikanth → Shaeed	"So accuracy and cost are clear. Shaeed will show what happens with fewer labels and on harder sentences."

Timing and rehearsal plan

Rehearsal	Goal	How
1: read aloud	know the content	read the sample text for each slide once, slowly
2: own words	natural speech	close the guide; explain each slide in your own words; time it
3: full run with a timer	7 to 10 minutes	use the rehearsal timer in the interactive study guide; target 9 minutes
4: with questions	calm answers	a friend asks 10 questions from section 14; practise the ABC method

IF YOU ARE RUNNING OUT OF TIME

Shorten slide 6 (implementation) and slide 8 (cost) to one sentence each. Never skip slide 11: the recommendation is what the professor remembers.

13. The storyline and the golden rules

The five-sentence storyline (memorise this)

1. **Problem:** Transformers are everywhere in NLP, but nobody tells a student with a laptop what they really give and cost.
2. **Gap:** the literature reports accuracy; cost and small-data behaviour are rarely measured together.
3. **Method:** ten models, one dataset, one laptop, same rules.
4. **Results:** Transformers are clearly more accurate and need far fewer labels, at 20 to 2,000 times the inference cost.
5. **So what:** reuse a ready-made model first, fine-tune a small Transformer second, keep TF-IDF for extreme cost limits.

A 30-second opening you can say

> *"Good morning. We are Shashikanth and Shaeed. Today's NLP systems are built on Transformers, but a student with a normal laptop wants to know: how much better are they than simple methods, and what do they cost? We compared ten sentiment classifiers on one laptop and measured accuracy, cost and the number of labels needed. Here is what we found."*

Golden rules

Do		Do not	
Say one number per slide, slowly	✓	Read the slide word for word	✗
Explain with one real sentence or analogy	✓	Say "proves" (say "shows" or "suggests")	✗
Say "our results suggest" for close differences	✓	Claim attention "explains" the model	✗
State limits yourself, before they are asked	✓	Compare 88.8 and 91.1 as if the gap were certain	✗
Look at the audience, point at the chart	✓	Hide a limitation or guess a number	✗
Hand over cleanly to your partner	✓	Talk over your partner or interrupt	✗

How to explain any chart in three steps

1 Axes
"On this axis is accuracy, on this one the number of labels."

2 Pattern
"The green line is always above the blue one; the gap is largest on the left."

3 Meaning
"So pre-training matters most when labels are scarce."

Honest wording table

Instead of...	Say...
"Transformers are always better."	"In our experiment the larger Transformers were clearly more accurate; the tiny ones were not."
"The attention map proves understanding."	"The map illustrates where information can flow."
"Our model is the best."	"The ready-made DistilBERT was best; ours, trained on 8,000 labels, was close and significantly above the classical models."
"We don't know."	"We did not test that; we would expect X because Y, and we would test it by Z."

14. Questions you may get, and how to answer

THE ABC METHOD FOR ANY QUESTION

Acknowledge ("Good question."), give a **B**rief answer in one or two sentences, **C**onnect it to evidence ("…as the learning curve on slide 9 shows"). If you do not know: "We did not test that. We would expect X because Y, and we would check it by Z." Never guess a number.

Question	Beginner-friendly answer
What is the aim of the project?	To analyse the Transformer architecture and to measure experimentally how much better it is than classical and recurrent methods for sentiment classification, including its cost, data needs and robustness.
Why sentiment analysis and why SST-2?	It is a common real task with published results for every model family, and it contains hard cases (negation, contrast) that test understanding of context. SST-2 is the standard benchmark for it.
What is a Transformer in simple words?	A network where every word looks at all other words at once (self-attention) instead of reading in order. That allows fast parallel training and very large pre-training.
What is pre-training versus fine-tuning?	Pre-training teaches general English from huge unlabelled text, once. Fine-tuning then teaches our specific task with a few thousand labelled sentences. That is why 8,000 labels were enough for 88.8%.
Why did you use the validation set as the test set?	GLUE does not publish test labels. We use the official validation set as held-out test data and never use it for learning; a separate development set of 2,000 training sentences is used to choose the best epoch.
Why only a CPU?	It is what students and many small deployments really have. The limit is that GPU timing would differ and the ratios between models might change; we state this as a limitation.
Which model is best and by how much?	The ready-made DistilBERT-SST2 (91.1%). Our fine-tuned DistilBERT (88.8%) is 5.8 points above the best classical model (82.9%), a significant difference (McNemar $p < 0.001$).
Is 88.8% versus 91.1% a real difference?	Their 95% intervals overlap (86.6 to 90.7 and 89.1 to 92.9), so we do not claim a clear difference. The ready-made model is probably a little better, which is plausible because it was fine-tuned on the full training set.
Why did the BiLSTM not beat TF-IDF?	It was trained from scratch without pre-training, so it must learn language and task from the same labels. It reached 82.8% against 82.9%; it is better on negation but not overall. This agrees with the Lithuanian study.
Why was BERT-tiny so weak?	It has only 4.4 million parameters, so its capacity is small. Size matters: tiny 76.6%, small 83.7%, DistilBERT 88.8% with the same training recipe.

14. Questions (continued)

Question	Beginner-friendly answer
What does the learning curve show?	With 500 labels BERT-small beats TF-IDF by 14.1 points; with 4,000 it matches the classical model trained on 65 k. Pre-training makes the model data-efficient.
What is the cost of using a Transformer?	20 to 2,000 times slower per sentence and much larger. On a laptop CPU that is about 40 ms per sentence for DistilBERT, which is acceptable for most applications.
What is a confidence interval?	A range showing how much the accuracy could change if we had a different sample of test sentences; we get it by resampling the test set many times (bootstrap).
What is McNemar's test?	A test for two models on the same sentences: it looks only at the sentences where exactly one model is right and asks whether the split is more uneven than chance would give.
How reproducible is your work?	All stages are scripts with a fixed seed (42), every model saves its predictions, and run_all.sh reproduces everything in about 2.5 to 3 hours. One limit: only one seed, so run-to-run variation is not measured.
Explain self-attention.	Each word makes a query, every word offers a key and a value. The match between query and keys, turned into weights by softmax, tells how much of each word's value to take. The result is a mix of the words that matter for that word.
Why divide by √d in the attention formula?	Dot products of long vectors become large and make the softmax almost 0 or 1, which stops learning. Dividing by √d keeps the scores moderate.
What are your limitations?	One dataset and language, one seed, a small test set (872), CPU-only timing, and a single fine-tuning size (8,000). We state them and describe how to remove each.
What would you do next?	Repeat with several seeds, add GPU timing, apply the same procedure to Lithuanian (LitLat BERT, XLM-RoBERTa) and other datasets, and add an error analysis of sentences all models get wrong.
What did each of you do?	Prepare this together beforehand. Suggested honest split in the paper: Shashikanth the design, the fine-tuning program, the Transformer theory and the practical part; Shaeed the introduction, the literature review, the classical and BiLSTM baselines, the error analysis and conclusions.

More questions with answers (54 in total, with a quiz mode) are in the interactive page `Project_Map_and_Interview_Prep.html`.

15. Day-of checklist and the one-page cheat sheet

The day before

- ☐ Two full timed rehearsals (target 9 minutes, never above 10)
- ☐ Agree who answers which kind of question
- ☐ Read the paper's introduction and conclusions once more
- ☐ Charge the laptop; copy files to a USB stick

Bring and open

- ☐ Presentation (PPTX) and a PDF copy as a backup
- ☐ The paper (PDF) open at the results and the architecture pages
- ☐ The interactive study guide and project map (work offline)
- ☐ Water, a pen, a watch for timing

During the talk

- ☐ Slow start; one clear sentence for the question
- ☐ Eye contact; do not read the slides
- ☐ Hand over with the prepared lines
- ☐ End with the recommendation and “thank you”

During questions

- ☐ Listen to the end; repeat the question if unclear
- ☐ ABC method; short answers
- ☐ Let your partner help if the question is theirs
- ☐ Be honest about limits

One-page cheat sheet

Item	Value
Task	binary sentiment, SST-2
Splits	65,349 / 2,000 / 872
Models	10: 3 TF-IDF, 1 BiLSTM, 3 fine-tuned, 3 ready-made
Best (ready-made)	DistilBERT-SST2 91.1%
Best trained by us	DistilBERT 88.8% (8,000 labels, 2.6 h)
Best classical	TF-IDF + SVM 82.9% (2 s, 0.06 ms)
BiLSTM	82.8%
Zero-shot	81.8%, no labels
500-label gap	BERT-small 75.9% vs TF-IDF 61.8%
Negation drop	SVM 8.4, DistilBERT-SST2 4.1 points
Cost	20 to 2,000× slower; DistilBERT ~40 ms per sentence
Hardware	i5-8365U, 4 cores, 16 GB, no GPU, seed 42
Test noise	about ±2.4 points on 872 sentences

OUR MOTTO

Explain it. Measure it. Make it useful for everyone.

GLOSSARY AND NEXT STEPS

16. Glossary

Accuracy
share of test sentences classified correctly

Attention
each word looking at other words and taking a weighted mix of their content

Backpropagation
working out how each weight should change to reduce the error

Bag of words
counting words and ignoring their order

BERT
a Transformer encoder pre-trained by guessing hidden words

BiLSTM
an LSTM that reads a sentence both forwards and backwards

[CLS]
a special token whose vector summarises the whole sentence for classification

Confidence interval
a range showing how much a result could wobble because of the test sample

DistilBERT
a smaller, faster BERT made by distillation (67 M parameters)

Fine-tuning
training a pre-trained model briefly on your own labelled task

Latency
time to classify one sentence

Learning curve
accuracy as the number of training labels grows

LSTM
a recurrent network with gates that keep long-term memory

Macro-F1
average of the F1 scores of the two classes

McNemar's test
paired test comparing two models on the same sentences

Pre-training
learning general language from huge unlabelled text

Ready-made model
a public model already trained for a task

SST-2
movie-review sentences labelled positive or negative

Self-attention
attention among the words of the same sentence

TF-IDF
word weights that favour rare, informative words

Token
a piece of text (word or word part) the model reads

Transformer
a network based on self-attention that processes all words at once

Zero-shot
using a model for a task it was not trained on, with no examples

Where to practise next

Material	Use it for
Project_Study_Guide.html	test your own sentences on all ten models; real attention maps; quiz; rehearsal timer
Project_Map_and_Interview_Prep.html	the clickable project flow and 54 interview questions with a quiz mode
Transformers_in_NLP_Presentation.pptx	the 11 slides with speaker notes
Transformers_in_NLP_Project_Paper.pdf	the full written work (25 pages)
Literature_Gap_Explained.pdf	the gap in the literature, with theory and an example
Transformer_Encoder_Decoder_Flow.pdf	how the encoder and decoder work, with real numbers

A LAST THOUGHT

You do not need to know everything. You need to explain clearly what you did, say honestly what the numbers show, and admit calmly what you did not test. That is what a professor wants to see.